

Heat Transfer to a Fluid in a Stirred Tank

Pre-Lab Plan

Unit Operations Lab **1**

January 21, 2020

Commented [DL1]: 30.5/35

Group #1, Tuesday, AM section

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1. Experimental Objective (5pts)

The objective of this experiment is to study a selection of steady state heat transfer processes. The emphasis of this study will be to measure the heat transfer coefficient between the water and the inside vessel wall of the continuously stirred tank. Parameters of the study will be manipulated to observe the dependence of the heat transfer coefficient on fluid properties, impeller speeds, and in the presence of baffles.

2. Experimental

The apparatus and the materials and supply of this experiment are introduced below and briefly explained their usage.

2.1 Apparatus (5 pts)

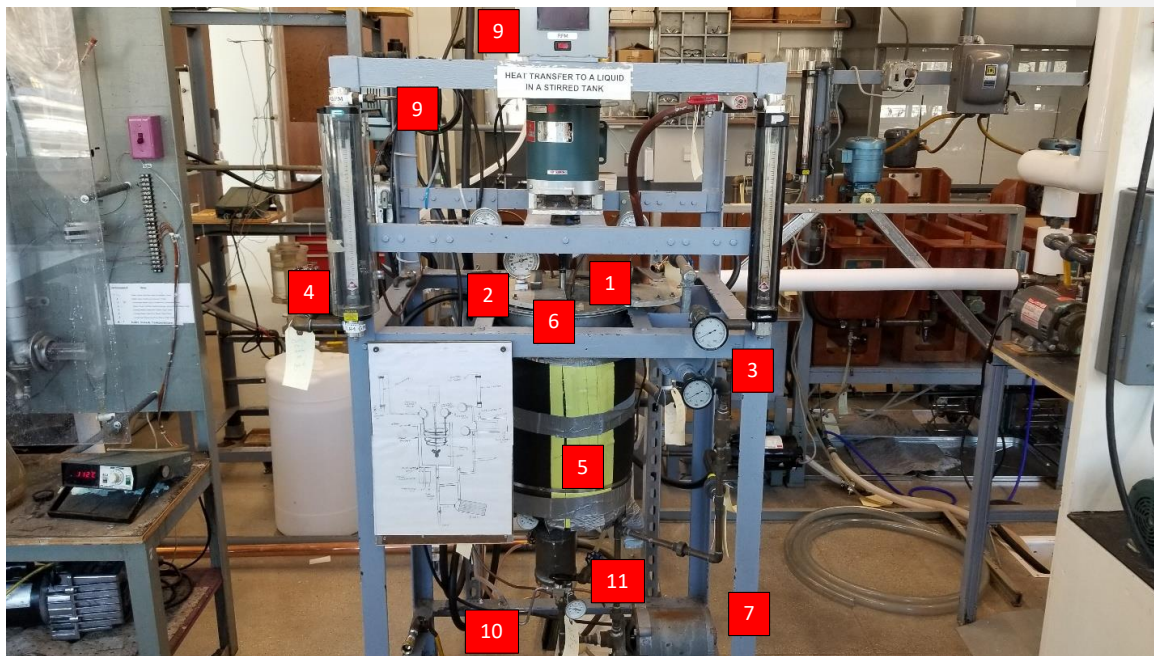


Figure 1: Apparatus used during the Heat Transfer experiment.

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Commented [DL3]: You can choose not only the steady state, but also the un-steady state.

Commented [DL4]: The main objective of this experiment is to measure the heat transfer coefficient between the water and the vessel wall. Additionally, it is required to examine the dependence of the measured heat transfer coefficient on the impeller speed, the presence of baffles, and the fluid properties that are highly dependent on temperature.

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A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

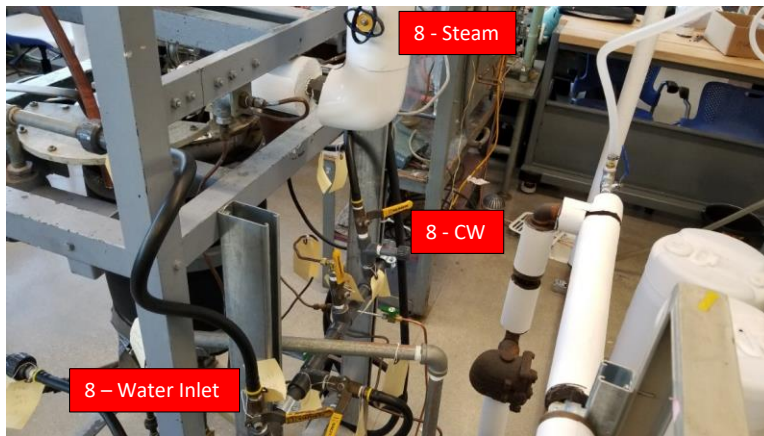


Figure 2: Back-end valves used during the Heat Transfer experiment.

Table 1. List of apparatus in the heat transfer unit with their description

1	Internal Cooling Coil	The internal cooling coil is used in order to regulate the temperature of the water in the CST
2	Dial Thermometers	The dial thermometers are used in order to see the inlet and outlet temperature of the cooling coil.
3	Heat exchanger	The heat exchanger is used to regulate the temperature of the water if the cooling coil is not able to cool the water to the desired temperature. The gauge is used to regulate the flow of water to the heat exchanger.
4	Flow Controller	The flow controller is used in order to control the amount of cold water that is being run through the cooling coil.
5	Steam Jacket	The steam jacket is wrapped around the reactor in order to rapidly heat the water inside of it.
6	Impeller	The Impeller is used to stir the water to ensure the temperature is uniform throughout the entirety of the volume of water.
7	Pump	Is used in order to pump the vessel water to the heat exchanger so that it can be additionally cooled to the desired temperature if needed.
8	Valves	There are several valves throughout the setup that are used to redirect the water either in or out of the CST so that it can be cooled properly. 8 - CW is the cooling water valve 8 - Steam valve for Steam Jacket 8 - Water Inlet valve for HE

Commented [DL6]: Include all gauges and rotameters.

Commented [DL7]: Unit and the maximum capacity.

9	RPM Controller/Meter	The RPM controller is used to change the speed of the impeller. The RPM Meter shows how fast the impeller is spinning.
10	Water Drain	This is used to expel all the water from the system throughout and at the end of the experiment.
11	Condensate Collect	This is where the condensation from the steam can be collected

2.2 Materials and Supplies (5 pts)

Table 2: Description of materials and supplies used during this experiment.

1	Water	Water is the main component used within the system. It is used in the cooling coil and in the tank.
2	Steam	Used as the main heating source in the CST. From house supply, available at 110 °C and 11 psig
3	Safety Goggles	Necessary to prevent hot liquids from entering your eyes.
4	Heat Resistant Gloves	Needed when controlling certain valves in order to prevent contact with the hot pieces of equipment from the steam.

2.3 Experimental Plan (10 pts)

General operating procedure:

- 1.) Fill the tank with water until the water level is two inches from the top of the tank.
- 2.) Turn on the impeller and adjust the speed to greater than 600 RPM
- 3.) Turn on the cooling water to the cooling coils by opening the control valve fully.
- 4.) Turn on the cooling water on the tube side of the shell and to the heat exchanger.
- 5.) Ensure that the outlet valve of the discharge line of the pump is fully closed and the suction and bypass valves are completely open.
- 6.) Turn on the pump and allow it to run for a few seconds while the discharge valve is closed.
- 7.) Open the discharge valve SLOWLY and close the bypass valve. This achieves a certain circulation rate. Ensure that the circulating liquid is fed back to the tank.
- 8.) Now SLOWLY open the inlet steam valve after turning on the cooling water to the steam condenser.
- 9.) When it has reached steady state, record all relevant properties needed for this experiment.

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Include the graduated cylinder and the stopwatch.

Commented [DL9]: Good, but could be altered at different days. Check those parameters before the experiment.

Commented [DL10]: 8.5/10

Commented [DL11]: Be specific.

Commented [DL12]: Be specific.

Commented [DL13]: What's the criteria of the steady state?

Commented [DL14]: List all parameters need to be measured.

Study using the unbaffled stirred tank

*Make sure the baffles are taken out of the tank

- 1.) Use a constant steam flowrate to the vessel, an impeller speed of 600 RPM, with water flowing through the cooling coils, and with the heat exchanger flowrate such that the temperature in the vessel is maintained at about 65 °C.
- 2.) Record measurements once it has reached steady state (i.e. temperatures, impeller speed, pressure of steam, and flowrates)
- 3.) Repeat everything from step 1 and 2 using an impeller speed of 800 and 1000 RPM.
- 4.) Repeat steps 1 and 2 using two different heat exchanger flowrates at 1000 RPM. This is done so that two different vessel temperatures are obtained. The two temperatures goals are 75 °C and 85 °C in the vessel. The flowrates cannot be determined beforehand but must be done experimentally.

Commented [DL15]: State the undeterminable flowrate as the step 4).

Commented [DL16]: Good.

Commented [AZ17]: Monitor 1000 RPM as it may be a bit high and cause the water to splash out of the vessel

Study using the baffled stirred tank

- 1.) Put the baffles in the stirred tank. Make sure they are secured and that the lid fully closes.
- 2.) Use a constant steam flowrate to the vessel, an impeller speed of 600 RPM, with water flowing through the cooling coils, and with the heat exchanger flowrate such that the temperature in the vessel is maintained at about 65 °C.
*Note: Make sure the open part of the baffle faces the direction of the flowing water in the tank.
- 3.) Record measurements once it has reached steady state (i.e. temperatures, impeller speed, pressure of steam, and flowrates)
- 4.) Repeat everything from step 1 and 2 using an impeller speed of 800 and 1000 RPM.
Repeat steps 1 and 2 using two different heat exchanger flowrates at 1000 RPM. This is done so that two different vessel temperatures are obtained. The two temperatures goals are 75 °C and 85 °C in the vessel. The flowrates cannot be determined beforehand but must be done experimentally.

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Project Safety Evaluation (10 pts)

Project Title: Heat Transfer to a Fluid in a Stirred Tank
Date: 1/21/2020

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	Yes/No	
Potential electrical hazards?	Yes	There are electrical hazards in this experiment which are the mixer, pump, and the RPM reader. Mixing this electronic equipment with the water in this experiment could lead to shock
Potential mechanical hazards?	Yes	A spinning impeller and a running motor can cause pinch points. This can potentially cause bodily harm.
Potential pressure hazards?	Yes	The potential pressures hazard can happen if the temperature of the vessel is not controlled and the steam piping could have a pressure increase.
Potential temperature hazards?	Yes	There is <u>are</u> multiple hot spots in the system like the vessel, and hot streams.
Hazardous/toxic chemicals involved?	No	N/A
Gloves required?	Yes	Good heat resistant gloves are needed to handle hot spots of the system.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water CAS #: 7732-18-5
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	110	The temperature inside the vessel is not expected to reach the max temperature, this is the max temperature of the steam
Pressure (psig)	11	The pressure of the steam.
<i>Safety Controls</i>	<i>Amount</i>	<i>Description</i>
Chill water on/off Valve	2	Controls the chill water flowrate that is used to cool down the vessel.
Drain Valve	1	Controls the amount of water inside the vessel.
Steam Valves	1	Controls the flowrate of steam

Commented [AZ20]: yes

Commented [DL21]: Use a wider range about 120 degree Celsius.

Commented [AZ22]: Pump and impeller switches

I have read relevant background material for the Unit Operations Laboratory entitled: "Heat Transfer to a Fluid in a Stirred Tank" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Diego Martinez	1/21/2020
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